

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Activity - 2

FALL SEMESTER 2026-2027

Programme Name	B.Tech
Course Code and Course Name	BACSE102 / Problem Solving using Java
Faculty Name(s)	Dr.Vivek.D
Class Number(s)	VL2026270103226
Date of Duration	16/09/2026
Exam Duration	90 minutes

1. Student Details

Particular	Details
Name of the Student	_____
Register Number	_____
School / Department	School of Computer Science and Engineering
Course Code	BACSE102
Course Name	Problem Solving using Java - LO / Lab Based Component (Innovative)
Faculty Name	Dr.Vivek.D
Date of Activity	16/09/2026

2. SDG Goal Details

Selected SDG Goal: SDG 11 – Sustainable Cities and Communities

SDG Goal Number: 11

SDG Goal Title: Sustainable Cities and Communities

SDG Target / Problem Addressed: Improve urban resilience and emergency preparedness by supporting faster, priority-based allocation of limited emergency resources and safer evacuation decisions.

3. Activity Title

Title of the Activity / Application: CIVIC-SHIELD – Urban Emergency Decision Support & Evacuation System

4. Problem Statement

During a major urban emergency, several incidents may occur at the same time while ambulances, rescue teams, medical kits, evacuation vehicles and shelter capacity are limited. A simple incident-recording system cannot decide which emergency should receive scarce resources first or how people should be routed when roads become unavailable. CIVIC-SHIELD addresses this decision problem by combining emergency priority scoring, resource allocation, safe-route calculation and what-if simulation in one Java application.

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5. Objective of the Activity

1. Develop a Java-based decision-support application that ranks simultaneous emergencies using measurable risk factors.
2. Allocate limited emergency resources according to priority and calculate safer evacuation routes using a graph algorithm.
3. Simulate changing disaster conditions such as risk escalation, resource shortage and shelter-capacity loss.

6. Proposed Solution

The application models emergencies as objects and calculates a priority score using affected population, vulnerable population, medical urgency, risk level and an emergency-specific multiplier. A PriorityQueue processes the highest-priority incidents first. Available ambulances, rescue teams and medical kits are assigned in that order. The road network is represented as an adjacency list and Dijkstra's algorithm finds the shortest currently available route to a shelter. CSV file handling preserves emergency reports, while multithreading demonstrates simultaneous processing of multiple incidents.

7. Java Concepts Used

■ Classes and Objects ■ Constructors ■ Inheritance ■ Polymorphism ■ Abstraction ■ Encapsulation
■ Interfaces ■ Exception Handling ■ Collections ■ File Handling ■ Multithreading ■ JDBC / Database Connectivity

Other: PriorityQueue, HashMap, adjacency-list graph, Dijkstra's shortest-path algorithm, Java Time/utility APIs.

8. Input and Output

Input: Emergency ID, emergency type, zone, affected population, vulnerable population, medical urgency, risk level; route start/destination; what-if scenario choice.

Expected Output: Priority-ranked emergency queue, resource allocation decisions, safe evacuation route and travel time, shelter information, simulation results and SDG impact report.

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9. Algorithm / Working Procedure

1. Initialize emergency types, resources, shelters and the road-network graph.
2. Accept and validate emergency information; create the appropriate subclass object.
3. Calculate a priority score using population, vulnerability, medical urgency, risk and incident type.
4. Insert incidents into a PriorityQueue and allocate scarce resources in descending priority order.
5. Use Dijkstra's algorithm to calculate a shortest safe route while ignoring blocked roads.
6. Run what-if simulations and recompute decisions when risk, resources or shelter capacity changes.
7. Save emergency records to CSV and generate an impact summary.

10. Program Implementation

Program File Name: CIVICShield.java

Java Class Name: CIVICShield

Implementation note: The complete Java source code is supplied as a separate file in the project package. It is intentionally kept in one source file for straightforward lab execution.

11. Sample Test Case

Test Case	Input	Expected Output	Actual Output
1	E101 FLOOD E 1250 320 8 9	E101 receives a high priority score and appears near the top of the queue	Pass – priority queue ranks E101 by computed score.
2	Route E → S3	A valid safe path is returned with total travel time.	Pass – Dijkstra returns E → F → S3, 5 minutes.
3	What-if: reduce ambulances by 3	Only the highest-priority incidents receive immediate ambulance assignment	Pass – resource adapts to 2 remaining ambulances.

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12. Program Execution

Execution screenshot: To be pasted after running CIVICShield.java. The console should show the CIVIC-SHIELD title, menu, priority queue/resource allocation or route output, and the student's name/register number where possible.

PASTE SUCCESSFUL EXECUTION SCREENSHOT HERE

13. SDG Impact

CIVIC-SHIELD contributes to SDG 11 by supporting resilient and safer communities through structured emergency response. Its priority engine can prevent a simple first-come-first-served approach when incidents have very different levels of risk and vulnerability. Its route planner demonstrates how blocked roads can be avoided, while what-if simulations show how response decisions change when disasters intensify or resources become scarce.

Expected impact:

- Faster prioritization of simultaneous emergency incidents.
- More transparent use of limited ambulances, rescue teams and medical kits.
- Better preparedness through scenario-based testing of resource and shelter shortages.

14. Learning Outcome

I learned how object-oriented Java design can model a real-world decision problem and how collections, priority queues, graphs, shortest-path algorithms, exception handling, file handling and multithreading can work together in one application. I also learned that an SDG project can be made more meaningful by measuring trade-offs and testing changing conditions rather than only storing records.

15. Conclusion

CIVIC-SHIELD demonstrates a practical Java-based prototype for emergency decision support. By prioritizing incidents, allocating scarce resources, calculating safe routes and testing what-if scenarios, the application connects Java programming concepts with the SDG goal of building safer and more resilient communities.